Yuyang Wang

Electrical & Computer Engineering
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Appointment

University of Connecticut — Storrs, CT, USA
Assistant Professor, Department of Electrical and Computer Engineering — 2025–Present

Columbia University in the City of New York — New York, NY, USA
Postdoctoral Research Scientist, Columbia Nano Initiative — 2021–2025

Education

University of California, Santa Barbara — Santa Barbara, CA, USA
Ph.D. in Electrical and Computer Engineering — 2018–2021

University of California, Santa Barbara — Santa Barbara, CA, USA
M.S. in Electrical and Computer Engineering — 2015–2018

Tsinghua University — Beijing, China
B.Eng. in Electronic Engineering — 2011–2015

Publication

Book Chapters

B1 **Y. Wang**, X. Meng, M. Lipson, A. L. Gaeta, and K. Bergman, "Co-packaged optics (CPO) for communications," in *Optical Fiber Telecommunications VIII*, A. E. Willner, Ed., chapter manuscript with the volume editor; book publication scheduled for March 1, 2027. Elsevier, 2027

Journal Articles

J1 B. George, **Y. Wang**, W. Wang, Z. Wu, X. Meng, M. Ghobadi, and K. Bergman, "RADICAL: Reconfiguration-Aware Direct-Connect AI Cluster using Spatial-and-Wavelength Selective Switching," *IEEE/Optica Journal of Optical Communications and Networking*, 2026, In press.

J2 Z. Wu, **Y. Wang**, and K. Bergman, "SiPAM: Silicon Photonic Accelerated Memory Pooling Architecture," *IEEE Micro*, pp. 1–10, 2026. doi 10.1109/MM.2026.3670009 Featured Article

J3 R. Parsons, K. Jang, **Y. Wang**, A. Novick, A. M. Smith, C. C. Tison, Y. Gebregiorgis, V. Deenadayalan, M. van Niekerk, L. Carpenter, T. Ngai, G. Leake, D. Coleman, X. Meng, S. Preble, M. L. Fanto, K. Bergman, and A. Rizzo, "Highly uniform thermally undercut silicon photonic devices in a 300 mm CMOS foundry process," *Scientific Reports*, vol. 15, no. 1, p. 29 906, Aug. 2025. doi 10.1038/s41598-025-14480-4

J4 R. Parsons, A. Oh, J. Robinson, S. Wang, M. Cullen, K. Jang, A. James, **Y. Wang**, and K. Bergman, "Foundry-enabled wafer-scale characterization and modeling of silicon photonic DWDM links," *Nanophotonics*, vol. 14, no. 27, pp. 5363–5374, 2025. doi 10.1515/nanoph-2025-0439

J5 J. Robinson, R. Parsons, **Y. Wang**, K. Jang, X. Meng, and K. Bergman, "Silicon photonic DWDM micro-resonator link initialization under fabrication variation," *Optics Express*, vol. 33, no. 26, pp. 53 700–53 713, Dec. 2025. doi 10.1364/OE.577064

J6 S. Wang, **Y. Wang**, S. Sanyal, R. Parsons, X. Ji, Y. Okawachi, X. Meng, M. Lipson, A. Gaeta, and K. Bergman, "Automated tuning of a ring-assisted MZI-based interleaver for Kerr frequency combs," *Optics Letters*, vol. 50, no. 2, p. 698, Jan. 2025. doi 10.1364/OL.546772

J7 **Y. Wang**, S. Wang, R. Parsons, S. Sanyal, V. Gopal, A. Novick, A. Rizzo, M. Lipson, A. L. Gaeta, and K. Bergman, "Co-Designed Silicon Photonics Chip I/O for Energy-Efficient Petascale Connectivity," *IEEE Transactions on Components, Packaging and Manufacturing Technology*, vol. 15, no. 8, pp. 1581–1591, Aug. 2025. doi 10.1109/TCPMT.2024.3492189 2025 IEEE Transactions on Components, Packaging and Manufacturing Technology Best Paper Award recipient

J8 Z. Wu, L. Yuan Dai, **Y. Wang**, S. Wang, and K. Bergman, "Flexible silicon photonic architecture for accelerating distributed deep learning," *Journal of Optical Communications and Networking*, vol. 16, no. 2, A157, Feb. 2024. doi 10.1364/JOCN.497372

J9 A. James, A. Rizzo, **Y. Wang**, A. Novick, S. Wang, R. Parsons, K. Jang, M. Hattink, and K. Bergman, "Process Variation-Aware Compact Model of Strip Waveguides for Photonic Circuit Simulation," *Journal of Lightwave Technology*, pp. 1–14, 2023. doi 10.1109/JLT.2023.3238847

J10 A. Novick, A. James, L. Y. Dai, Z. Wu, A. Rizzo, S. Wang, **Y. Wang**, M. Hattink, V. Gopal, K. Jang, R. Parsons, and K. Bergman, "High-bandwidth density silicon photonic resonators for energy-efficient optical interconnects," *Applied Physics Reviews*, vol. 10, no. 4, p. 041 306, Dec. 2023. doi 10.1063/5.0160441

J11 **Y. Wang**, P. Sun, J. Hulme, M. A. Seyedi, M. Fiorentino, R. G. Beausoleil, and K.-T. Cheng, "Energy Efficiency and Yield Optimization for Optical Interconnects via Transceiver Grouping," *Journal of Lightwave Technology*, vol. 39, no. 6, pp. 1567–1578, Mar. 2021. 10.1109/JLT.2020.3039489

J12 Z. Zhang, R. Wu, **Y. Wang**, C. Zhang, E. J. Stanton, C. L. Schow, K.-T. Cheng, and J. E. Bowers, "Compact Modeling for Silicon Photonic Heterogeneously Integrated Circuits," *Journal of Lightwave Technology*, vol. 35, no. 14, pp. 2973–2980, Jul. 2017. 10.1109/JLT.2017.2706721

Conference Proceedings

C1 M. Cullen, S. Sanyal, S. Wang, K. J. McNulty, **Y. Wang**, M. Lipson, A. L. Gaeta, and K. Bergman, "Reconfigurable Kerr-comb for enhanced link margins in >1 Tb/s direct-drive DWDM links," in *Conference on Lasers and Electro-Optics (CLEO)*, Charlotte, NC: Optica Publishing Group, May 2026, STh4A.1. 10.1364/CLEO_SI.2026.STH4A.1 **Highlighted**

C2 B. C. George, W. Wang, Z. Wu, **Y. Wang**, X. Meng, M. Ghobadi, and K. Bergman, "Reconfiguration-aware direct-connect AI cluster using spatial-and-wavelength selective switching," in *Optical Fiber Communication Conference (OFC)*, Optica Publishing Group, Mar. 2026, M3F.4. 10.1364/OFC.2026.M3F.4 **Top-Scored**

C3 S. Sanyal, M. Cullen, S. Wang, **Y. Wang**, Y. Okawachi, K. J. McNulty, M. Lipson, K. Bergman, and A. L. Gaeta, "High-power Kerr comb source for data communications," in *Optical Fiber Communication Conference (OFC)*, Optica Publishing Group, Mar. 2026, Th1F.5. 10.1364/OFC.2026.Th1F.5

C4 S. Wang, M. Cullen, T. Zypman, S. Sanyal, K. J. McNulty, **Y. Wang**, X. Meng, M. Lipson, A. L. Gaeta, and K. Bergman, "On-chip programmable MZI-based Fourier synthesizer for ultra-broadband Kerr-comb equalization," in *Optical Fiber Communication Conference (OFC)*, Optica Publishing Group, Mar. 2026, M1E.4. 10.1364/OFC.2026.M1E.4

C5 N. Nauman, J. Robinson, **Y. Wang**, K. Jang, X. Meng, and K. Bergman, "Photonic Analog-to-Digital Architecture for Accelerating Multiply-Accumulate Operations," in *Optical Fiber Communication Conference (OFC)*, Optica Publishing Group, Mar. 2025, W3D.5. 10.1364/OFC.2025.W3D.5

C6 R. Parsons, S. Sanyal, M. Cullen, **Y. Wang**, A. Novick, X. Ji, Y. Okawachi, X. Meng, M. Lipson, A. Gaeta, and K. Bergman, "Dispersion-Engineered Resonator-Based Interleaver Co-Designed with Kerr Comb Source," in *Optical Fiber Communication Conference (OFC)*, Optica Publishing Group, Mar. 2025, Tu2G.6. 10.1364/OFC.2025.Tu2G.6

C7 A. Rovinski, Y. Ou, C. Ou, D. Khilwani, **Y. Wang**, S. Wang, S. Lee, K. Bergman, A. Molnar, and C. Batten, "Scaling Co-Packaged Optical Interconnects Using Hybrid 2.5D/3D Integration," in *IEEE International Symposium on Circuits and Systems (ISCAS)*, May 2025, pp. 1–5. 10.1109/ISCAS56072.2025.11043946

C8 **Y. Wang**, S. Wang, S. Sanyal, N. Nauman, R. Parsons, J. Robinson, M. Hattink, K. Jang, A. Novick, K. J. McNulty, X. Meng, M. Lipson, A. L. Gaeta, and K. Bergman, "Order-Preserving Channel Calibration of Kerr Comb–Driven Microresonator-Based DWDM Link," in *Optical Fiber Communication Conference (OFC)*, Optica Publishing Group, Mar. 2025, W1E.1. 10.1364/OFC.2025.W1E.1

C9 A. Novick, M. Hattink, A. Rizzo, **Y. Wang**, V. Gopal, S. Wang, R. Parsons, and K. Bergman, "Integrated Photonic Resonant Modulator-Based Equalization and Optimization for DWDM," in *Optical Fiber Communication Conference (OFC)*, San Diego, CA, USA: Optica Publishing Group, 2024, W1K.5. 10.1364/OFC.2024.W1K.5

C10 S. Wang, **Y. Wang**, X. Meng, K. Hosseini, T. T. Hoang, and K. Bergman, "Automated Tuning of Ring-Assisted MZI-Based Interleaver for DWDM Systems," in *Optical Fiber Communication Conference (OFC)*, San Diego, CA, USA: Optica Publishing Group, 2024, Th1A.3. 10.1364/OFC.2024.Th1A.3

C11 **Y. Wang**, S. Wang, R. Parsons, A. Novick, V. Gopal, K. Jang, A. Rizzo, C.-P. Chiu, K. Hosseini, T. T. Hoang, S. Shumarayev, and K. Bergman, "Silicon Photonics Chip I/O for Ultra High-Bandwidth and Energy-Efficient Die-to-Die Connectivity," in *IEEE Custom Integrated Circuits Conference (CICC)*, Denver, CO, USA: IEEE, Apr. 2024, pp. 1–8. 10.1109/CICC60959.2024.10529018 **Invited**

C12 Z. Wu, R. Parsons, S. Wang, **Y. Wang**, and K. Bergman, "Wavelength Reconfigurable Transceiver for Multi-Interface Compute Accelerator Networks," in *Optical Fiber Communication Conference (OFC)*, San Diego, CA, USA: Optica Publishing Group, 2024, W4F.2. 10.1364/OFC.2024.W4F.2

C13 G. Michelogiannakis, Y. Arafa, B. Cook, L. Y. Dai, A.-H. Hameed Badawy, M. Glick, **Y. Wang**, K. Bergman, and J. Shalf, "Efficient Intra-Rack Resource Disaggregation for HPC Using Co-Packaged DWDM Photonics," in *IEEE International Conference on Cluster Computing (CLUSTER)*, Santa Fe, NM, USA: IEEE, Oct. 2023, pp. 158–172. 10.1109/CLUSTER52292.2023.00021

C14 S. Wang, A. Novick, A. Rizzo, R. Parsons, S. Sanyal, K. J. McNulty, B. Y. Kim, Y. Okawachi, **Y. Wang**, A. Gaeta, M. Lipson, and K. Bergman, "Integrated, Compact, and Tunable Band-Interleaving of a Kerr Comb Source," in *Conference on Lasers and Electro-Optics (CLEO)*, San Jose, CA: Optica Publishing Group, 2023, STh3J.6. 10.1364/CLEO_SI.2023.STh3J.6

C15 **Y. Wang**, A. Novick, R. Parsons, S. Wang, K. Jang, A. James, M. Hattink, V. Gopal, A. Rizzo, C.-P. Chiu, K. Hosseini, T. T. Hoang, and K. Bergman, "Scalable architecture for sub-pJ/b multi-Tbps comb-driven DWDM silicon photonic transceiver," in *Next-Generation Optical Communication: Components, Sub-Systems, and Systems XII*, G. Li, K. Nakajima, and A. K. Srivastava, Eds., San Francisco, United States: SPIE, Mar. 2023, p. 55. 10.1117/12.2649506 **Invited**

C16 **Y. Wang**, S. Wang, A. Novick, A. James, R. Parsons, A. Rizzo, and K. Bergman, "Dispersion-Engineered and Fabrication-Robust SOI Waveguides for Ultra-Broadband DWDM," in *Optical Fiber Communication Conference (OFC)*, San Diego, CA, USA: Optica Publishing Group, 2023, Th3A.4. 10.1364/OFC.2023.Th3A.4

C17 A. James, **Y. Wang**, A. Rizzo, and K. Bergman, "Flexible, Process-Aware Compact Model of Effective Index in Silicon Waveguides for Commercial Foundries," in *International Conference on Numerical Simulation of Optoelectronic Devices (NUSOD)*, Turin, Italy: IEEE, Sep. 2022, pp. 173–174. 10.1109/NUSOD54938.2022.9894784

C18 **Y. Wang** and K.-T. Cheng, "Traffic-Adaptive Power Reconfiguration for Energy-Efficient and Energy-Proportional Optical Interconnects," in *IEEE/ACM International Conference on Computer-Aided Design (ICCAD)*, Munich, Germany: IEEE, Nov. 2021, pp. 1–9. 10.1109/ICCAD51958.2021.9643475

C19 **Y. Wang**, J. Hulme, P. Sun, M. Jain, M. A. Seyedi, M. Fiorentino, R. G. Beausoleil, and K.-T. Cheng, "Characterization and Applications of Spatial Variation Models for Silicon Microring-Based Optical Transceivers," in *ACM/IEEE Design Automation Conference (DAC)*, San Francisco, CA, USA: IEEE, Jul. 2020, pp. 1–6. 10.1109/DAC18072.2020.9218608

C20 **Y. Wang** and K.-T. Cheng, "Task Mapping-Assisted Laser Power Scaling for Optical Network-on-Chips," in *IEEE/ACM International Conference on Computer-Aided Design (ICCAD)*, Westminster, CO, USA: IEEE, Nov. 2019, pp. 1–6. 10.1109/ICCAD45719.2019.8942146

C21 **Y. Wang**, M. A. Seyedi, J. Hulme, M. Fiorentino, R. G. Beausoleil, and K.-T. Cheng, "Bidirectional tuning of microring-based silicon photonic transceivers for optimal energy efficiency," in *Asia and South Pacific Design Automation Conference (ASP-DAC)*, Tokyo, Japan: ACM, Jan. 2019, pp. 370–375. 10.1145/3287624.3287649

C22 **Y. Wang**, L. Shao, M. A. Lastras-Montano, and K.-T. Cheng, "Taming Emerging Devices' Variation and Reliability Challenges with Architectural and System Solutions [Invited]," in *IEEE International Conference on Microelectronic Test Structures (ICMTS)*, Kita-Kyushu City, Fukuoka, Japan: IEEE, Mar. 2019, pp. 90–95. 10.1109/ICMTS.2019.8730924 Invited

C23 **Y. Wang**, M. A. Seyedi, R. Wu, J. Hulme, M. Fiorentino, R. G. Beausoleil, and K.-T. Cheng, "Energy-efficient channel alignment of DWDM silicon photonic transceivers," in *Design, Automation & Test in Europe Conference & Exhibition (DATE)*, Dresden, Germany: IEEE, Mar. 2018, pp. 601–604. 10.23919/DATE.2018.8342079

C24 R. Wu, M. A. Seyedi, **Y. Wang**, J. Hulme, M. Fiorentino, R. G. Beausoleil, and K.-T. Cheng, "Pairing of microring-based silicon photonic transceivers for tuning power optimization," in *Asia and South Pacific Design Automation Conference (ASP-DAC)*, Jeju, South Korea: IEEE, Jan. 2018, pp. 135–140. 10.1109/ASPDAC.2018.8297295

C25 R. Wu, **Y. Wang**, Z. Zhang, C. Zhang, C. L. Schow, J. E. Bowers, and K.-T. Cheng, "Compact modeling and circuit-level simulation of silicon nanophotonic interconnects," in *Design, Automation & Test in Europe Conference & Exhibition (DATE)*, Lausanne, Switzerland: IEEE, Mar. 2017, pp. 602–605. 10.23919/DATE.2017.7927057

C26 A. Ghofrani, M. A. Lastras-Montaño, **Y. Wang**, and K.-T. Cheng, "In-place Repair for Resistive Memories Utilizing Complementary Resistive Switches," in *International Symposium on Low Power Electronics and Design (ISLPED)*, San Francisco Airport, CA, USA: ACM, Aug. 2016, pp. 350–355. 10.1145/2934583.2934590

C27 C. Xu, F. X. Lin, **Y. Wang**, and L. Zhong, "Automated OS-level Device Runtime Power Management," in *International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS)*, Istanbul, Turkey: ACM, Mar. 2015, pp. 239–252. 10.1145/2694344.2694360

Under Review

U1 X. Meng, **Y. Wang**, Z. Wu, and K. Bergman, "Silicon photonics for data centers," *Nature Materials*, 2026 Invited Perspective

U2 R. Parsons, J. Robinson, **Y. Wang**, U. D. Dave, X. Meng, M. Lipson, and K. Bergman, "Silicon photonic integration of DWDM and mode-division multiplexing for highly parallel optical interconnects," *Optics Express*, 2026

Mentoring

U : undergraduate; **M** : master's; **D** : doctoral

Waseem Faheem **D** Ph.D. Student, University of Connecticut — Spring 2026 – Present

Zahin Zeal **D** Ph.D. Student, University of Connecticut — Fall 2026 – Present

Harvansh Pelia **U** Undergraduate Student, University of Connecticut — Spring 2026 – Present
Honors Thesis Supervision

Willow Pichardo **U** Undergraduate Student, University of Connecticut — Spring 2026 – Present

Uyen Bao Pham **U** Undergraduate Student, University of Connecticut — Spring 2026 – Present

Khang "James" Le U Undergraduate Student, University of Connecticut — Spring 2026 – Present

Abidur Rahman U Undergraduate Student, Wayne State University — Summer 2024
Summer Undergraduate Research Experience (SURE) Program of Columbia Engineering

Teaching

U : undergraduate; G : graduate

University of Connecticut — Storrs, CT, USA

– Instructor	ECE 2001: Electrical Circuits U	Spring 2026, 2027
– Project Advisor	ECE 4901: Senior Design U	Fall 2025 – Present

Columbia University in the City of New York — New York, NY, USA

– Guest Lecturer	ELEN 9404: Seminar in Lightwave Communications G	Spring 2024
– Guest Lecturer	ELEN 9403: Seminar in Photonics G	Spring 2023

University of California, Santa Barbara — Santa Barbara, CA, USA

– Teaching Assistant	ECE 153B: Sensor & Peripheral Interface Design U	Winter 2019

Service

Guest Editor IEEE Solid-State Circuits Magazine — 2027
Spring 2027 Issue: Silicon Photonics

Technical Program Committee Optical Fiber Communication Conference (OFC) — 2027
N2: Optics and Photonics for Data Center and Computing Applications

Workshop Organizing Committee Optical Fiber Communication Conference (OFC) — 2027
Breaking the Tyranny of Copper: Optical Scale-Up and the Path to Disaggregation

Topical Chair IEEE Photonics Society Summer Topicals Meeting Series (SUM) — 2026
Electronic-Photonic Systems at Scale — Architecture, Methodology, and Going 3D (EP3D)

Technical Program Committee IEEE Hot Interconnects Symposium (HotI) — 2026

Technical Program Committee IEEE Chiplet Workshop: Standards, Circuits and Systems — 2025
Chiplet Interconnects and Packaging

Ad-hoc Reviewer

- *Nature Photonics,* Springer Nature
- *Nature Nanotechnology,* Springer Nature
- *Nature Communications,* Springer Nature
- *npj Nanophotonics,* Springer Nature
- *Communications Engineering,* Springer Nature
- *APL Photonics,* American Institute of Physics
- *Photonics*, MDPI
- *ACM Transactions on Design Automation of Electronic Systems,* ACM
- *Journal of Lightwave Technology,* IEEE/Optica

- *IEEE Photonics Journal,* IEEE
- *Journal of Selected Areas in Communications,* IEEE
- *Journal on Emerging and Selected Topics in Circuits and Systems,* IEEE
- *IEEE Transactions on Computers,* IEEE
- *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems,* IEEE
- *IEEE Transactions on Very Large Scale Integration (VLSI) Systems,* IEEE
- *IEEE Access,* IEEE

Award and Honor

Best Paper Award Recipient IEEE Transactions on Components, Packaging and Manufacturing Technology 2025
"Co-Designed Silicon Photonics Chip I/O for Energy-Efficient Petascale Connectivity"

Presentation

Invited Talk IEEE Photonics Conference (IPC) Denver, CO, USA
From DWDM Links to Photonic Fabrics: Link-Budget-Driven Co-Design for Heterogeneous Computing Nov. 2026

Seminar Department of Physics, University of Connecticut Storrs, CT, USA
Integrated Photonics Systems in the Package: Beyond Massive Parallel Data Movement Mar. 2026

Seminar Department of ECE, University of Connecticut Storrs, CT, USA
Photonics-Empowered Computing: From Efficient to Functional Data Movement Feb. 2025

Seminar School of ECEE, Arizona State University Tempe, AZ, USA
Photonics-Empowered Computing: From Efficient to Functional Data Movement Feb. 2025

Seminar Department of ECE, University of Minnesota Minneapolis, MN, USA
Photonics-Empowered Computing: From Efficient to Functional Data Movement Feb. 2025

Presentation SRC TECHCON Austin, TX, USA
Scalable Kerr-Comb Driven DWDM Silicon Photonics Chip I/O Sep. 2024

Seminar School of EECS, Pennsylvania State University University Park, PA, USA
Co-Designing Photonics for Heterogeneously Integrated Systems Apr. 2024

Invited Talk SPIE Photonics West San Francisco, CA, USA
Scalable Architecture for Sub-pJ/b Multi-Tbps Comb-Driven DWDM Silicon Photonic Transceiver Jan. 2023

Poster Ph.D. Forum, ACM/IEEE Design Automation Conference (DAC) Online Virtual Event
Design and Optimization of Variation-Aware Runtime-Reconfigurable Optical Interconnects Jun. 2020

Invited Talk Optical/Photonic Interconnects for Computing Systems (OPTICS) Workshop Dresden, Germany
Optimal Pairing and Non-Uniform Channel Alignment of Microring-Based Transceivers for Comb Laser–Driven DWDM Silicon Photonics Mar. 2018

Seminar Department of ECE, Hong Kong University of Science and Technology Hong Kong SAR, China
Variation-Aware Modeling and Design of Silicon Photonic Systems Jan. 2018

Poster Optical/Photonic Interconnects for Computing Systems (OPTICS) Workshop Lausanne, Switzerland
Variation-Aware Modeling and Design of Nanophotonic Interconnects Mar. 2017